BrightLight Tutorials – Data Analytics

SQL Fundamentals

Exercise 3: SQL CASE Statements

Question 1:

Classify each product by price:

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Answers:

```
SELECT product_name, price,
CASE
   WHEN price > 1000 THEN 'Expensive'
   WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
   ELSE 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Question 2:

Label each order:

- 'High Value' for orders ≥ 1000
- 'Medium Value' for 500–999.99
- 'Low Value' for orders < 500

Answer:

```
SELECT customer_name, amount,
CASE
   WHEN amount >= 1000 THEN 'High Value'
   WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
   ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Question 3:

Categorize employee position:

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

Answer:

```
SELECT emp_name, department, salary,
CASE
   WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
   WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
   ELSE 'Staff'
END AS position_level
FROM employees;
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Question 4:

Assign a letter grade:

- ≥ 90: 'A'
- 80–89: 'B'
- 70–79: 'C'
- 60–69: 'D'
- < 60: 'F'

Answer:

```
SELECT student_name, score,
CASE
   WHEN score >= 90 THEN 'A'
   WHEN score BETWEEN 80 AND 89 THEN 'B'
   WHEN score BETWEEN 70 AND 79 THEN 'C'
   WHEN score BETWEEN 60 AND 69 THEN 'D'
   ELSE 'F'
END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Question 5:

Label delivery performance:

- ≤ 30 mins: 'Fast'
- 31–60 mins: 'On Time'
- 60 mins: 'Late'

Answer:

```
SELECT delivery_id, delivery_time_minutes,
```

```
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
    ELSE 'Late'
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Question 6:

Convert priority to labels:

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

Answer:

```
SELECT issue_type, priority,
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    ELSE 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login Issue	1	Low
Server Down	3	Hight
Slow System	2	Medium
Email Error	2	Medium
Password Reset	1	Low

Question 7:

Calculate attendance % and classify:

- ≥ 90% → 'Excellent'
- 75–89% → 'Good'
- < 75% → 'Needs Improvement'

Answer:

```
SELECT student_id,
(days_present * 100.0 / total_days) AS attendance_percentage,
CASE
   WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'
   WHEN (days_present * 100.0 / total_days) BETWEEN 75 AND 89 THEN 'Good'
   ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Question 8:

Label stock status:

- 0 → 'Out of Stock'
- 1–5 → 'Low Stock'
- 5 → 'In Stock'

Answer:

```
SELECT product_id, stock_qty,
CASE
   WHEN stock_qty = 0 THEN 'Out of Stock'
   WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
```

```
    ELSE 'In Stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Question 9:

Classify by size:

- ≥ 25 → 'Large'
- 10–24 → 'Medium'
- < 10 → 'Small'

Answer:

```
SELECT subject, enrolled_students,
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    ELSE 'Small'
END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Question 10:

Apply discount flag:

- If payment_method = 'Cash' and amount ≥ 200 → 'Eligible for Discount'

● Otherwise → 'Not Eligible'

Answer:

```
SELECT payment_id, payment_method, amount,

CASE

   WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'

   ELSE 'Not Eligible'

END AS discount_eligibility

FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible For Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible For Discount